

INNOVA JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION 2  
in preparation for General Certificate of Education Advanced Level  
**Higher 2**

CANDIDATE  
NAME

CLASS

INDEX NUMBER

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**MATHEMATICS**

**9740/02**

Paper 2

**22 September 2009**

**3 hours**

Additional Materials:      Answer Paper  
                                        Cover Page  
                                        List of Formulae (MF15)

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**READ THESE INSTRUCTIONS FIRST**

**Do not open this booklet until you are told to do so.**

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. You may use a soft pencil for any diagrams or graphs.

**Do not use staples, paper clips, highlighters, glue or correction fluid.**

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

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This document consists of **7** printed pages and **1** blank page.



## Section A: Pure Mathematics [40 marks]

- 1 The curve  $C$  has equation

$$y = \frac{x^2 + 9x + 16}{x + 2}.$$

- (i) Find the equations of the asymptotes of  $C$ . [2]
- (ii) Find, leaving your answers in exact form, the range of values of  $k$  for which the equation  $k = \frac{x^2 + 9x + 16}{x + 2}$  has no real roots. [4]
- (iii) Sketch  $C$ . Show, on your diagram, the equations of the asymptotes, the coordinates of the stationary points, and the coordinates of the points of intersection of  $C$  with the  $x$ - and  $y$ - axes. [3]

- 2 (a) The plane  $\Pi$  is given by the vector equation

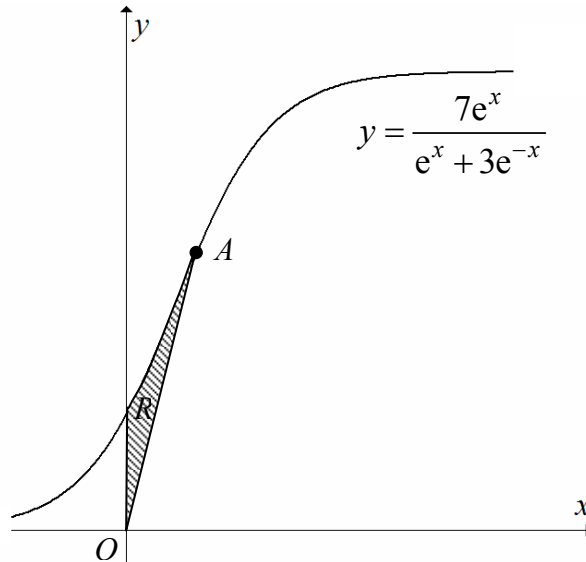
$$\mathbf{r} \cdot \mathbf{n} = -14.$$

Given that  $|\mathbf{n}| = \frac{7}{6}$  units, find the shortest distance from the origin  $O$  to the plane  $\Pi$ . [2]

- (b) Relative to the origin  $O$ , the position vectors of points  $A$  and  $B$  are  $\mathbf{a}$  and  $\mathbf{b}$  respectively. Given that angle  $AOB$  is  $90^\circ$ , show that the position vector of the foot of the perpendicular from  $O$  to  $AB$  is

$$\mathbf{a} + \frac{|\mathbf{a}|^2}{|\mathbf{a}|^2 + |\mathbf{b}|^2}(\mathbf{b} - \mathbf{a}). \quad [5]$$

- 3 The diagram shows the region  $R$  bounded by the curve  $y = \frac{7e^x}{e^x + 3e^{-x}}$ , the  $y$ -axis, and the line  $OA$ , where  $A$  is the point on the curve when  $x = \ln 2$ .



- (i) By means of the substitution  $u = e^x$ , find the exact value of

$$\int_0^{\ln 2} \frac{e^x}{e^x + 3e^{-x}} dx. \quad [4]$$

- (ii) Calculate the exact area of the region  $R$ . [2]
- (iii) Find the numerical value of the volume of the solid formed when  $R$  is rotated through 4 right angles about the  $x$ -axis. [3]
- (iv) Find the equation of the curve obtained when the curve  $y = \frac{7e^x}{e^x + 3e^{-x}}$  is translated 5 units in the negative direction of the  $y$ -axis, followed by a stretch parallel to the  $x$ -axis with a scale factor of 3. [2]

- 4 (i) Show that, for  $z \neq -2$ ,

$$\sum_{j=0}^3 \left( -\frac{z}{2} \right)^j = \frac{16-z^4}{16+8z}. \quad [2]$$

- (ii) The non-zero roots of the equation

$$\sum_{j=0}^3 \left( -\frac{z}{2} \right)^j - 1 = 0$$

are denoted by  $z_1$  and  $z_2$ , where  $\arg(z_2) > 0$ . Find the complex numbers  $z_1$  and  $z_2$ , leaving your answers in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ . [4]

- (iii) On an Argand diagram, with origin  $O$ , mark the points  $A$  and  $B$  representing the complex numbers  $z_1$  and  $z_2$  respectively. The locus of points representing the complex number  $w$  has equation  $|w| = k$  and passes through  $A$  and  $B$ . State the value of  $k$  and sketch the locus  $|w| = k$  in the same Argand diagram. [2]

- (iv) The tangents to the locus  $|w| = k$  at the points  $A$  and  $B$  intersect at point  $P$ . Find  $s$ , where  $s$  is the length of  $OP$ . [2]

- (v) The point  $Q$  represents the variable complex number  $w$  satisfying both the inequalities  $k \leq |w| \leq s$  and  $\arg(z_1) \leq \arg(w) \leq \arg(z_2)$ . Show clearly on the Argand diagram in (iii) the locus of the point  $Q$ . Hence find the exact area of the region in which the point  $Q$  lies. [3]

**Section B: Statistics [60 marks]**

- 5 Find the number of 4-letter code-words that can be formed from the thirteen letters of the word MEDITERRANEAN such that the first letter is E and the last letter is R. [3]
- 6 A gigantic Christmas tree contains a large number of light bulbs which are continually being switched on and off. Individual lights fail at random times, independently, and each day the lights are inspected and any failed lights are replaced. The number of lights that fail in any one-day period has a Poisson distribution with mean  $\mu$ . Find the exact value of  $\mu$  such that the probability that at least one light bulb has to be replaced on a particular day is 0.3. [3]
- 7 The Headmaster of a school requires a sample of 40 students from a cohort of 2000 to complete a students' perception survey.
- (i) He numbers each student from 1 to 2000 and generates 40 distinct random numbers,  $a_1, a_2, \dots, a_{40}$ , from a computer. However, due to the computer settings, the numbers generated are in three decimal places, such that  $0.001 \leq a_i \leq 1.000$ ,  $i = 1, 2, \dots, 40$ . He then multiplies these numbers by 2000 to obtain 40 integers, which he uses to select a sample of 40 students. State, with a reason, whether the sample obtained is a simple random sample. [2]
- (ii) Describe briefly how the Headmaster can select a systematic random sample. [2]
- 8 In a game, three containers have the following contents.
- A jar contains 10 red balls and 15 blue balls.
- A white box contains 4 mint chocolates and 8 almond chocolates.
- A black box contains 7 milk chocolates and 3 almond chocolates.
- One ball is drawn at random from the jar. If the ball is red, a chocolate will be drawn at random from the white box. If the ball is blue, a chocolate will be drawn at random from the black box. Events  $X$  and  $Y$  are defined as follows.
- $X$ : A red ball is drawn from the jar.
- $Y$ : The second item drawn is an almond chocolate.
- Find, giving each of your answers as an exact fraction in its lowest terms,
- (i)  $P(Y)$ , [2]
- (ii)  $P(X \cap Y)$ , [2]
- (iii)  $P(X | Y)$ . [2]
- Determine whether  $X$  and  $Y$  are independent events. [2]

- 9** Durians are sold at \$7 per kilogram. The mass of a randomly chosen durian is normally distributed with mean 1.5 kg and standard deviation 0.2 kg. Mangoes, which are sold in bags of five, are priced at \$2 per kilogram. The mass of a randomly chosen mango is normally distributed with mean 0.4 kg and standard deviation 0.05 kg. The masses of durians and mangoes may be assumed to be independent of each other.

- (i) Find the probability that a randomly chosen bag of mangoes has a selling price exceeding \$4.20. [3]
- (ii) Find the probability of the event that both a randomly chosen durian has a selling price exceeding \$10 and a randomly chosen bag of mangoes has a selling price exceeding \$4.20. [2]
- (iii) Find the probability that the total selling price of 2 durians and 3 bags of mangoes is less than \$33.50. [4]

[The mass of an empty bag is assumed to be negligible.]

- 10** (a) An inventor has developed a new, energy-efficient lawn mower engine. He claims that the engine will run continuously for 5 hours on a single gallon of regular gasoline. A random sample of 15 engines is tested and the mean run time on a single gallon of regular gasoline is found to be 290 minutes with a standard deviation of 20 minutes. Test, at the 5% level of significance, whether the inventor has overstated the mean run time.

State any assumptions necessary in order for the test to be valid. [5]

- (b) A random sample of 50 packets of ‘Nilo Bar’ snacks is weighed and the mass,  $x$  grams, is recorded. The results are summarised by

$$\Sigma(x - 150) = -250, \quad \Sigma(x - 150)^2 = 3000.$$

The population mean mass is  $\mu$  grams. A test is carried out at the 1% significance level with the following hypotheses:

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

Given that  $H_0$  is rejected in favour of  $H_1$ , find the set of possible values of  $\mu_0$ . [5]

- 11** A horticulturist carries out a research looking into how the yield of tomatoes is affected by the amount of fertilizer applied to tomato plants. Ten tomato plants of the same variety were selected at random and treated, weekly, with a solution in which  $x$  grams of fertilizer was dissolved in a fixed quantity of water. The yield,  $y$  kilograms, of tomatoes was recorded. The results are given in the table.

| Plant | A   | B   | C   | D   | E   | F   | G   | H   | I   | J   |
|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $x$   | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 4.5 | 5.0 | 5.5 |
| $y$   | 2.1 | 4.4 | 5.8 | 6.6 | 7.0 | 7.1 | 7.3 | 7.7 | 7.8 | 7.9 |

- (i) Calculate the product moment correlation coefficient between  $x$  and  $y$ , and explain whether your answer suggests that a linear model is appropriate. [2]
- (ii) Draw a scatter diagram for the data and explain why it may be consistent with a model of the form  $y = a + b \ln x$ . [2]
- (iii) Calculate least squares estimates of  $a$  and  $b$  for the model  $y = a + b \ln x$ . [2]
- (iv) Calculate the product moment correlation coefficient for the transformed data and comment on its value. [2]

Use a regression line to give the best estimate that you can of the amount of fertilizer used on a plant of the same variety when the yield of tomatoes is 6 kilograms. [2]

- 12** (a) Fifty random observations are taken from a binomial distribution  $B(20, 0.2)$ . Find the probability that the sample mean lies between 3.8 and 4.2. [3]
- (b) There is a spread of a disease caused by a new type of virus in a certain community and the probability,  $p$ , of a randomly chosen person being infected with the disease is 0.25. Find the probability that, in a random sample of 10 people chosen from this community, at least 4 are infected with the disease. [3]

After some time, the virus mutates. As a result, the value of  $p$  increases to 0.65. By using a suitable approximation, find the probability that, in a random sample of 100 people, there are at least 60 but not more than 80 who are infected with the disease. [4]

The virus mutates further such that  $p$  is now 0.98. By using a suitable approximation, find the probability that, in a random sample of 200 people, at least 196 are infected with the disease. [3]